



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**THIRD YEAR SECOND SEMESTER EXAMINATION FOR
THE DEGREE OF BACHELOR OF SCIENCE IN CHEMISTRY WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

SCH 304: GROUP THEORY

Date: 8th July, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer ANY FOUR Questions.



INSTRUCTIONS

Time: 3 hours

Answer any our questions

1. a). Give the meaning of the following terms as applied in group theory?

(i) Chiral molecule

(1 mark)

(ii) Polar molecule

(1 mark)

b). Use group theory to determine whether the following molecules are chiral or polar:

(i) CH_2CF_2 (*cis*)

(ii) OCS

(iii) $\text{C}_6\text{H}_5\text{Br}$

(7.5 marks)

c). A molecule belongs to the point group C_{2v} . A central atom in this molecule uses the orbitals $2s$, $2p_x$, $2p_y$ and $2p_z$ in covalent bonding with the other atoms in the molecule.

(i) List down the symmetry operators of the point group.

(2 marks)

(ii) Use these wave functions as bases to generate irreducible representations for the point group

(6 marks)

2. Consider the molecule $\text{Cr}(\text{NH}_3)_4\text{ClF}$ (with the Cl and F atoms on opposite sides).

a) Sketch the structure of this molecule.

(1 mark)

b) Determine the symmetry operations possible for this molecule.

(7 marks)

c) What is meant by the terms 'collinear axes'? Identify any for this molecule.

(2 marks)

d) What is meant by the term class as applied to symmetry operations of a molecule?

Identify the classes of this molecule.

(4.5 marks)

e) Identify the point group for this molecule.

(1 mark)

- f) Test the law of commutativity by considering any two symmetry elements apart from the common element held by every molecule. (2 marks)

3. Given the following species: BBr_2F , $\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2$ (the fluorides lying on opposite sides), *trans*- $\text{C}_2\text{H}_2\text{Cl}_2$, *cis*- $\text{C}_2\text{H}_2\text{Cl}_2$ and XeCl_4

- a) Determine the total symmetry for each of these species (12 marks)
- b) Identify the principal axis if any for each species (2.5 marks)
- c) Identify key symmetry elements for each if any for each species (3 marks)

4. a) Give the relationship between each of the following terms as applied to molecules in group theory:

- (i) Principal axis and horizontal plane
- (ii) Dihedral plane and vertical plane
- (iii) Molecular plane and horizontal plane

(5 marks)

b). Use ammonia molecule structure to illustrate what is meant by 'improper operation' (3 marks)

c) What is meant by:

(i) Representation of a point group? (1 mark)

(ii) The term 'order' as applied to in a point group? (1 mark)

d) The following symbols are used to label some point groups:

(i). T_d

(ii). I_h

(iii). C_i

(iv). $D_{\infty h}$

(v) C_{nv}

What condition(s) must be satisfied by a molecule to be classified under each of this point groups? (7.5 marks)

5. a) Symmetry operations on a molecule lead to indistinguishable configurations

(i) What is meant by the expression a 'mathematical group'? (1 mark)

(ii) State the four rules that must be obeyed by a set of members in a group. (4 marks)

b) A molecule can undergo a number of symmetry operations carried through symmetry elements to get indistinguishable configurations depending on the concerned molecule. List down the different types of symmetry operations and their corresponding elements and their symbols. (12.5 marks)



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**THIRD YEAR SECOND SEMESTER EXAMINATION FOR
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INFORMATION TECHNOLOGY**

MAIN CAMPUS

SCH 307: INSTRUMENTATION IN CHEMICAL ANALYSIS

Date: 13th July, 2018

Time : 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO



Useful information:

$$A = \epsilon Cnl$$

$$A = \log I_0/I$$

$$E = \mu H/I$$

$$E = h\nu$$

$$HETP = L/N$$

$$N = 16(t_R/w)^2$$

1. Question 1 (30 marks)

- a) Differentiate between:
- UV-visible spectroscopy and fluorescence spectroscopy (2 marks)
 - Selectivity and sensitivity (2 marks)
 - Flame Atomic absorption spectroscopy and Inductively coupled plasma atomic emission spectroscopy (2 marks)
- b)
- What is the purpose of a potentiostat in voltammetry? (1 mark)
 - What is the advantage of using Hg as a working electrode in polarography over other metals? (2 marks)
 - State two disadvantages of polarography. (2 marks)
- c) Mass spectroscopy is a powerful, widely used analytical technique.
- On what principle is mass spectroscopy based? (1 mark)
 - State two applications of mass spectroscopy. (2 marks)
 - What is tandem mass spectroscopy? (1 mark)
- d) Nuclear Magnetic Resonance (NMR) spectroscopy is a widely used technique in synthetic chemistry, natural products chemistry and biochemistry, among other applications.
- What do you understand by the term Larmor frequency in nuclear magnetic resonance (NMR) spectroscopy? (1 mark)
 - Given the magnetic field strength H of a NMR spectroscope, the magnetic moment μ and spin quantum number I , derive an expression for Larmor frequency in terms of H , μ and I , assuming wave-like properties of the energy absorbed by the nuclei. (2 marks)
 - State two limitations of NMR spectroscopy? (2 marks)
- e)
- State two sources of radiation in infrared spectroscopic instruments (1 mark)
 - What are the different types of molecular vibrations that can occur when compounds are exposed to infrared electromagnetic radiation (4 marks)
- f) Differentiate among the following:
- Gas chromatography, high-performance liquid chromatography and electrophoresis (3 marks)
 - Isocratic elution and gradient elution in high-performance liquid chromatography (2 marks)

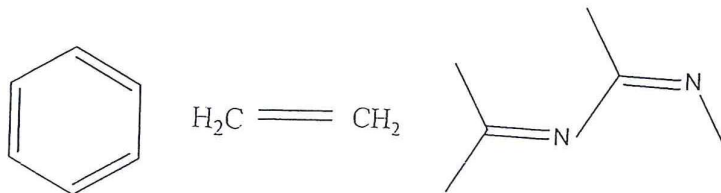
2. Question 2 (20 marks)

- a) Highlight the differences between coherent radiation and incoherent radiation. Give an example of each. (3 marks)
- b) Outline the main steps in the atomization process of a continuously introduced sample solution in atomic spectroscopy. (4 marks)

- c) Differentiate between
- Atomic Absorption Spectroscopy and Atomic Emission Spectroscopy (2 marks)
 - K series and L series in atomic x-ray fluorescence spectroscopy (2 marks)
 - Dispersive grating and Fourier transform IR spectrometers (2 marks)
- d)
- What is the purpose of a blank analysis in atomic emission spectroscopy? (1 mark)
 - State three ways through which plasmas can be formed in atomic emission spectroscopy (3 marks)
- e) A chromatographic peak has a retention time of 85 seconds. The base width of the peak is equivalent to 4.6 seconds by intersection of the peak with the baseline. Given that the column length is 800 cm, calculate the height equivalent of theoretical plates in terms of cm per plate (3 marks)

Question 3 (20 marks)

- a)
- Using a schematic diagram, show the main components of a double beam UV-visible spectrophotometer (3 marks)
 - Differentiate between bathochromic and hypsochromic shifts in UV-visible spectroscopy (2 marks)
- b) A student performs a set of UV-visible determinations on a set of unknown samples for an analyte known to have maximum absorbance at a wavelength of 500 nm. All of the student's results give absorbances of between 3 and 4.
- How reliable are these results? Explain. (1 mark)
 - What would you advise the student to do? (1 mark)
 - A UV-visible spectrophotometric analysis of a solution shows an absorbance of 0.52 at 520 nm. What percentage of radiation is transmitted? (2 marks)
 - A compound with a molar absorptivity of $3578 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$ exhibits an absorbance of 0.78 when placed in a 1 cm path length cuvette in a UV-visible spectrophotometer. Calculate the concentration of the compound. (2 marks)
- c) Arrange the three chromophores below in order of increasing absorbance. Explain. (3 marks)



d) Explain how phosphorescence spectroscopy occurs (3 marks)

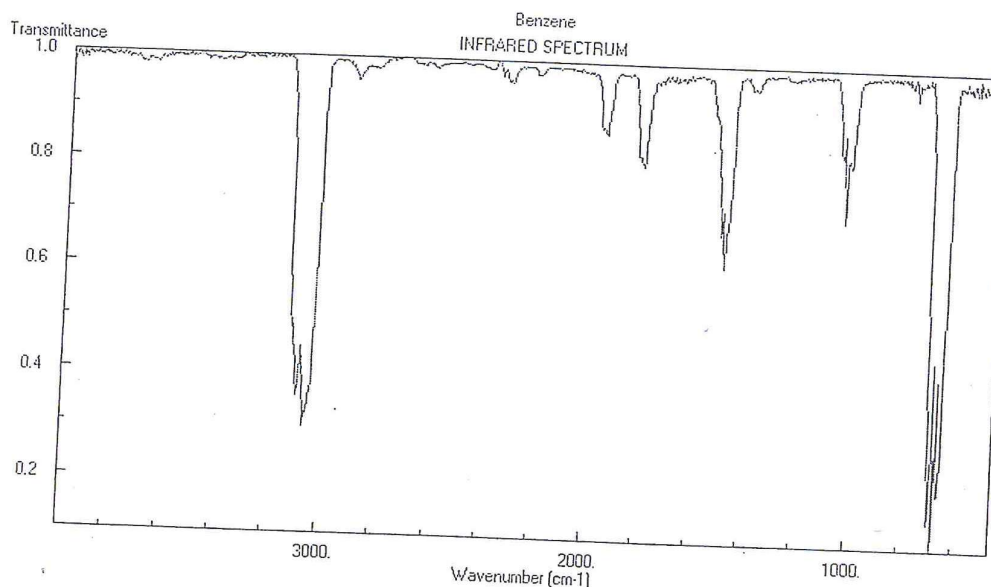
e) Explain the difference between x-ray auger-electron and x-ray fluorescence processes in x-ray spectroscopy (3 marks)

Question 4 (20 marks)

- a) What do you understand by the following terms in chromatography:
- i. Stationary phase (1 mark)
 - ii. Retention time (1 mark)
 - iii. Resolution (1 mark)

- b) Gas chromatography is one of the techniques commonly used. Explain how the electron capture detector works in Gas Chromatography (3 marks)
- i. What is band broadening? (1 mark)
 - ii. Explain four ways through which band broadening could occur (4 marks)

c) Shown below is the infrared absorption spectrum for benzene:



- i. The C-H stretch occurs at 3068 cm⁻¹. Calculate the corresponding wavelength in μm . (2 marks)
- ii. Calculate the energy of a mole of photons (in J) having this wavelength. (3 marks)
- iii. The vertical axis of this spectrum is given in transmittance. Estimate the transmittance of the C-H stretch and then calculate the corresponding absorbance of this stretch. (2 marks)
- iv. The C-H bend occurs at 1045 cm⁻¹. Which of the two - a C-H stretch or a C-H bend - is a higher energy transition? Explain your answer. (3 marks)



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INFORMATION TECHNOLOGY**

MAIN CAMPUS

SCH 311: ELECTROCHEMISTRY

Date: 8th July, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer ALL Questions.



SCH 311: ELECTROCHEMISTRY

INSTRUCTIONS:

Time: 3 Hours

Attempt **ALL** questions

Standard Reduction potential/periodic table attached

Physical Constants

$$R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}$$

$$F = 96,500 \text{ JVmole}^{-1}$$

$$N_A = 6.023 \times 10^{23} \text{ mol}^{-1}$$

$$\Delta H_f^\circ(\text{NH}_{3(g)}) = -46.11 \text{ kJmol}^{-1}$$

$$\Delta H_f^\circ(\text{H}_2\text{O}_{(l)}) = -285.83 \text{ kJmol}^{-1}$$

Q1. a) Explain the following expressions

(i) Oxidation half-reaction (1 mark)

(ii) reducing agent (1 mark)

(iii) standard reduction potential (1 mark)

b) Draw diagrams for the galvanic cells utilizing the following half-cells

Cathode

Anode

(i) Cl_2/Cl^- Br^-/Br (2 marks)

(ii) Ni^{2+}/Ni Al/Al^{3+} (2 marks)

(iii) Write balanced equations for the overall cell reactions (above)

(4 marks)

- c) (i) Write the cell conventional notation of a galvanic cell consisting of an

Al electrode placed in a 1.0M $\text{Al}(\text{NO}_3)_3$ solution and a Ag in a 1.0 M AgNO_3 solution. (1 mark)

(ii) Write the cell reaction that occurs under standard state condition in c)(i) above. (1 mark)

(iii) Calculate the e.m.f (E_{cell}°) of a cell that uses Ag/Ag^+ and Al/Al^{3+} half-cell reaction. (1 mark)

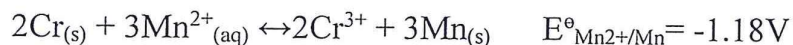
Q2. (a) (i) What is free energy? (1 mark)

(ii) What is the condition for spontaneity? (1 mark)

(iii) Derive the equation

$$E_{\text{cell}}^\circ = \frac{RT}{nF} \ln K \quad (2 \text{ marks})$$

(b) Consider the following reaction



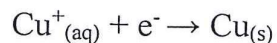
Calculate

(i) E_{cell}° (2 marks)

(ii) ΔG° (2 marks)

(iii) K (2 marks)

(c) Given that $E^\circ = 0.52\text{V}$ for the reduction

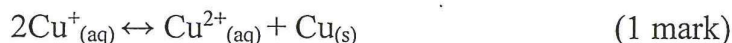


Calculate

(i) E_{cell}° (2 marks)

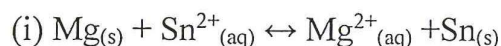
(ii) ΔG° (1 mark)

(iii) K, for the reaction at 25°C ,

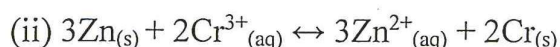


- Q3. a) (i) Derive the Nernst equation, and explain all the terms. (3 marks)
 (ii) What is the potential of a cell made up of Zn/Zn^{2+} and Cu/Cu^{2+} half-cells at 25°C of $[\text{Zn}^{2+}] = 0.25\text{M}$ and $[\text{Cu}^{2+}] = 0.15\text{M}$. (2 marks)

b) Calculate E° , E and ΔG for the following cell reactions



$[\text{Mg}^{2+}] = 0.045\text{M}$, $[\text{Sn}^{2+}] = 0.035\text{M}$



$[\text{Cr}^{3+}] = 0.010\text{M}$, $[\text{Zn}^{2+}] = 0.0085\text{M}$ (5 marks)

- c) (i) Calculate the standard potential of a cell consisting of the

Zn/Zn^{2+} half-cell and the SHE (1 marks)

(ii) What will the e.m.f of the cell be if $\text{Zn}^{2+} = 0.45\text{M}$, $P_{\text{H}_2} = 1.0\text{ atm}$ and $[\text{H}^+] = 1.8\text{M}$ (3 marks)

- Q4. a) (i) What is a fuel cell? (1 mark)

(ii) Using an example, explain how a fuel cell works. (3 marks)

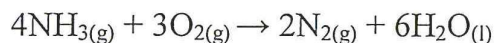
- b) (i) What is a concentration cell? (1 mark)

(ii) Briefly explain how a concentration cell operates. (2 marks)

(iii) Calculate the E°_{cell} of the following concentration cell.



c) Consider the oxidation of ammonia



(i) Calculate the ΔG° for the reaction (3 marks)

(ii) If this reaction were used in a fuel cell what would be the standard cell potential be? (2 marks)

- Q5. a) (i) What is an electrolytic cell? (1 mark)
- (ii) Explain how an electrolytic cell operates (2 marks)
- b) (i) It took 2.30 min using a current of 2.00A to plate out all of the silver from 0.250 L of a solution containing Ag^+ . What was the original concentration of Ag^+ in the solution? (3 marks)
- (ii) A solution containing a 3^+ metal ion is electrolyzed by a current of 5.0A for 10.0 min. What is the identity of the metal if 1.18 g of metal was plated out in this electrolysis? (4 marks)
- c) A solution at 25°C contains the following metal ions:
- | | |
|--|---------------------------|
| $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$ | $E^\circ = 0.80\text{V}$ |
| $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ | $E^\circ = 0.34\text{V}$ |
| $\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$ | $E^\circ = -0.76\text{V}$ |

Explain the order in which the metals will plate out, if this solution is electrolyzed? (4 marks)



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MAIN CAMPUS

SCH 307: INSTRUMENTATION IN CHEMICAL ANALYSIS

Date: 21st September, 2018

Time 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions.



SCH 307: INSTRUMENTATION IN CHEMICAL ANALYSIS

INSTRUCTIONS:

Answer QUESTION 1 (COMPULSORY) and any other TWO questions

QUESTION ONE [30 MARKS]

- a) State Beer-Lambert's Law. [2 marks]
- b) A solution containing 5 mg/100 ml of KMnO_4 (f.Wt = 158) had a transmittance of 36.4% when measured in a 1.05 cm cuvette at a wavelength of 525 nm. Calculate the molar absorptivity of the solution? [3 marks]
- c) Briefly explain the three deviations from Beer-Lambert's Law. [6 marks]
- d) (i) What do you understand by the term calibration of instrumental methods? [1 mark]
(ii) State three methods which are used to calibrate instrumental methods. [3 marks]
- e) Differentiate between the following terms;
(i) Sensitivity and selectivity [2 marks]
(ii) Detection limit and dynamic range {2 marks]
(iii) Fluorescence and phosphorescence [2 marks]
- f) (i) Define the following terms:
I. Spin-spin coupling {1 mark]
II. Coupling constant J [1 mark]
III. Equivalent hydrogens {1 mark]
(ii) A 300-MHz spectrophotometer records a proton that absorbs at a frequency of 2130 Hz downfield (deshielded) from TMS.
I. Determine its chemical shift [2 marks]
II. Predict this proton's chemical shift at 60-MHz {1 mark]

- g) List the two major wavelength selectors used in instrumental analytical methods. [2 marks]

QUESTION TWO [20 MARKS]

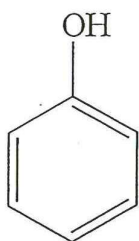
- a) Describe the basic principles of flame emission spectroscopy. [4 marks]
- b) An industrial chemistry student was given a fresh leaf sample and asked to determine the content of potassium in the sample. Describe the procedure the student used for the determination. [5 marks]
- c) Differentiate between “fuel-rich” and “fuel-lean” flames as used in AAS. [1 marks]
- d) (i) Explain using an example, the formation of refractory compounds during atomic absorption spectroscopic analysis. [1 marks]
- (ii) Explain how the formation of refractory compounds is prevented. [2 marks]
- e) The data below was obtained during the determination of iron in industrial waste stream.

Concentration (ppm)	Absorbance
0.00	0.00
1.00	0.183
2.00	0.346
3.00	0.546
4.00	0.727
Unknown (waste stream sample)	0.632

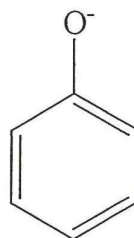
Determine the concentration of iron in the waste stream. [7 marks]

QUESTION THREE [20 MARKS]

- a) (i) What are the four sources of radiation applied in ultra-violet spectrophotometry? [2 marks]
- (ii) State the advantages of using double-beam ultra-violet spectrophotometers over single-beam units. [3 marks]
- (iii) Differentiate between the following terms as used in ultra-violet spectrophotometry:
- I. Hypsochromic shift and bathochromic shift [1 mark]
 - II. Chromophore and auxochrome. [1 mark]
- (iv) Predict which of the following structures will absorb at longer wavelength in ultra-violet spectrometry. Explain your answer. [2 marks]



A



B

- (v) Draw the energy level diagram for the following energy transitions in UV spectrophotometry.
 $\pi \rightarrow \pi^*$
 $n \rightarrow \pi^*$ [1 mark]
- b) (i) State the requirements for absorption of infrared radiation by a molecule. [2 marks]
- (ii) The wavenumber of the fifth overtone is 3000 cm^{-1} . Calculate the wavenumber of:
- I. The fundamental vibration frequency [2 marks]
 - II. The second overtone [2 marks]

(iii) Determine the expected number of IR absorption peaks of CO₂ molecule. [1 marks]

(iv) The IR spectrum of CO₂ shows two absorption peaks at 2350 cm⁻¹ and 667 cm⁻¹. Account for this observation. [3 marks]

QUESTION FOUR [20 MARKS]

- a) Define the following terms as used in chromatography. [1 mark]
- (i) Stationary phase [1 mark]
 - (ii) Mobile Phase [1 mark]
 - (iii) Retention time [1 mark]
- b) Distinguish between gas chromatography (GC) and high performance liquid chromatography (HPLC). [4 marks]
- c) List four requirements of pumps used in HPLC. [5 marks]
- d) Substances A and B have retention times of 16.40 and 17.63 minutes, respectively on a 30 cm column. An unretained species passes the column in 1.30 minutes. The peak widths for A and B are 1.11 and 1.21 minutes, respectively. Determine;
- (i) The column resolution [2 marks]
 - (ii) Average number of theoretical plates [4 marks]
 - (iii) The plate height [2 marks]